Chapter 1 - Introduction and overview

Course authors (Git file)

Section 1

Welcome

What we have done so far

- VHDL language introduction
- RTL Simulation with GHDL
- Synthesis for FPGA with Intel/Altera Quartus
- Basic circuits: Counter, Shiftregister, Statemachine
- Lab measurements with Terasic DE1 board

Section 2

Course overview

Chapter names

- 1 Introduction
- 2 OpenROAD tools
- 3 Verilog
- 4 OpenROAD first run
- 5 PDK
- 6 OpenROAD GUI
- 7 OpenROAD flow scripts
- 8 Tapeout

Get the course materials here:

Course materials:

https://github.com/fredowski/Course/tree/tha

- Look in the “tha” branch
- pdfs are in the “build” directory
- There might be daily updates during the course!

Additional course related links:

Master VLSI Wiki:

https://www.hs-augsburg.de/homes/beckmanf/dokuwiki/doku.php?id=msvsli_start

OpenROAD Flow Scripts:

https://github.com/fredowski/OpenROAD-flow-scripts

Duplicated content versus internet links

The course slides

- contain Links to the Internet for a lot of topics.
- do not contain duplicated content (or as less as possible).

This means:

- Follow the links and read there. It is important content for the course.
- The links are carefully curated. It's not spamming.
- Don't expect all the content beeing duplicated into the course slides.

Section 3

Open-source EDA for digital designs

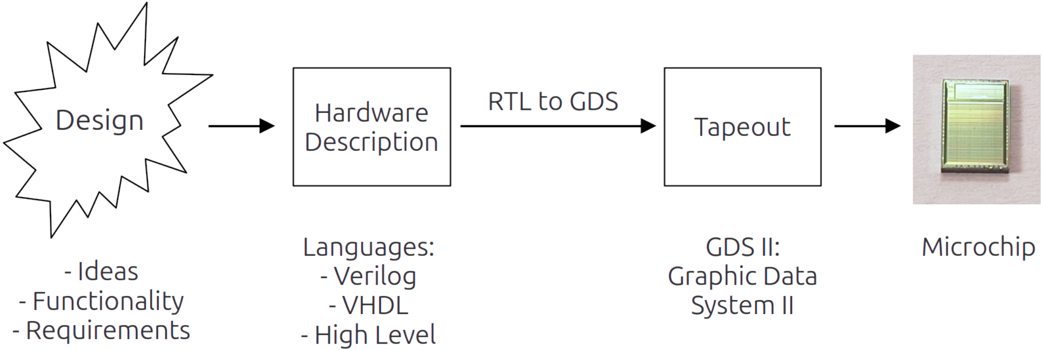
Digital designs

There are:

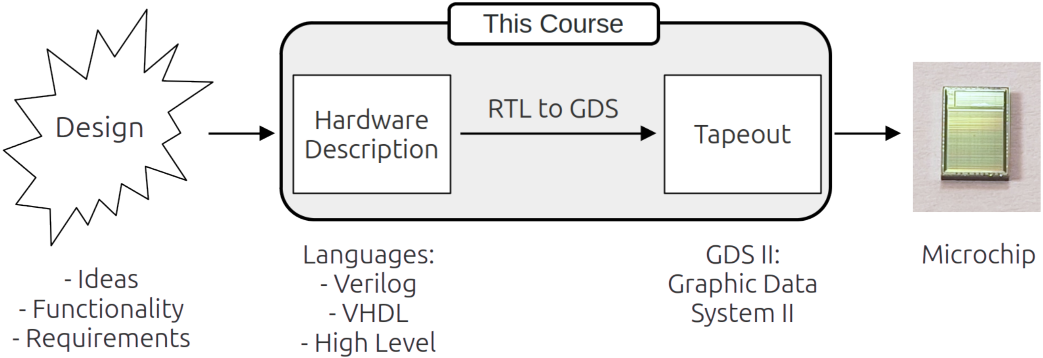
- **Digital designs** This course!
- Analog designs (Upcoming course)
- Mixed signal designs
- Artwork designs (i.e. Minimal Fab Contest)
  - https://github.com/mineda-support/Semicon2023-MinimalFab-Design-Contest
- Your fancy design?

From now on: This course means digital design, even if not mentioned everywhere again

From design to microchip



RTL to GDS toolchain



RTL: Register Transfer Level

In digital circuit design, **register-transfer level** (**RTL**) is a design abstraction which models a synchronous digital circuit in terms of the flow of digital signals (data) between hardware registers, and the logical operations performed on those signals.

Register-transfer-level abstraction is used in hardware description languages (HDLs) like Verilog and VHDL to create high-level representations of a circuit, from which lower-level representations and ultimately actual wiring can be derived. Design at the RTL level is typical practice in modern digital design.[1]

Unlike in software compiler design, where the register-transfer level is an intermediate representation and at the lowest level, the RTL level is the usual input that circuit designers operate on. In fact, in circuit synthesis, an intermediate language between the input register transfer level representation and the target netlist is sometimes used. Unlike in netlist, constructs such as cells, functions, and multi-bit registers are available.[2] Examples include FIRRTL and RTLIL.

Transaction-level modeling is a higher level of electronic system design.

RTL description [edit]

A synchronous circuit consists of two kinds of elements: registers (sequential logic) and combinational logic. Registers (usually implemented as D flip-flops) synchronize the circuit's operation to the edges of the clock signal, and are the only elements in the circuit that have memory properties. Combinational logic performs all the logical functions in the circuit and it typically consists of logic gates.

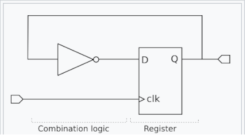


Figure 1: RTL (Screenshot from Wikipedia [1])

[1] https://en.wikipedia.org/wiki/Register-transfer_level

GDS: Graphic Data System (II)

GDSII

7 languages

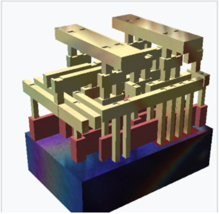
Article Talk Read Edit View history Tools

From Wikipedia, the free encyclopedia

GDSII stream format (**GDSII**), is a binary database file format which is the de facto industry standard for electronic design automation (EDA) data exchange of integrated circuit (IC) or IC layout artwork.[1] It is a binary file format representing planar geometric shapes, text labels, and other information about the layout in hierarchical form (two-dimensional/2D CAD file format). The data can be used to reconstruct all or part of the artwork to be used in sharing layouts, transferring artwork between different tools, or creating photomasks.

GDSII

Filename extension	.gds
Developed by	Calma
Initial release	1989; 36 years ago
Type of format	binary[1]
Free format?	no



History [edit]

GDS = Graphic Design System (see [GDS78])

Initially, GDSII was designed as a stream format used to control integrated circuit photomask plotting. Despite its limited set of features and low data density, it became the industry conventional stream format for transfer of IC layout data between design tools of different vendors, all of which operated with proprietary data formats.

It was originally developed by Calma for its layout design system, "Graphic Design System" ("GDS") and "GDSII".

Figure 2: GDS (Screenshot from Wikipedia [2])

[2] https://en.wikipedia.org/wiki/GDSII

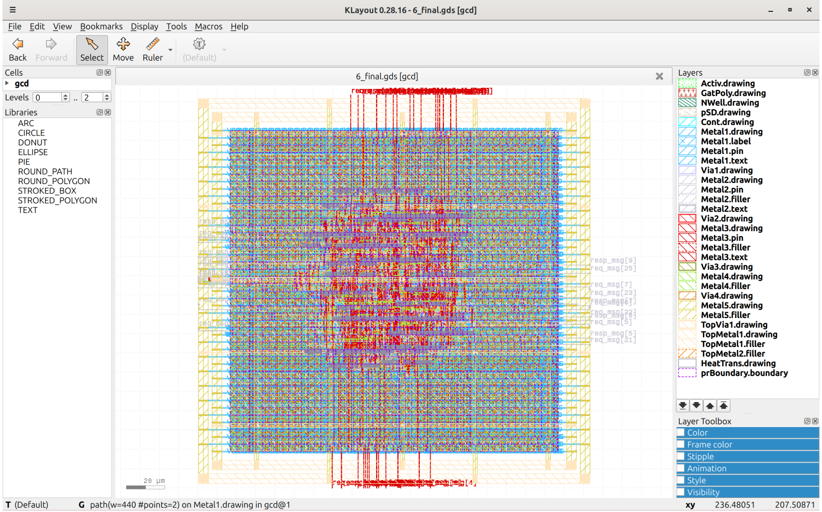
The GDS II Format (Specifications)

Here are two links about the structure, format and elements of GDS II. The links are for reference reasons. It is not strictly necessary to read or learn the GDS II format for this course. But it might help understanding.

https://boolean.klaasholwerda.nl/interface/bnf/gdsformat.html

https://www.rulabinsky.com/cavd/text/chapc.html

Example of a GDS in KLayout



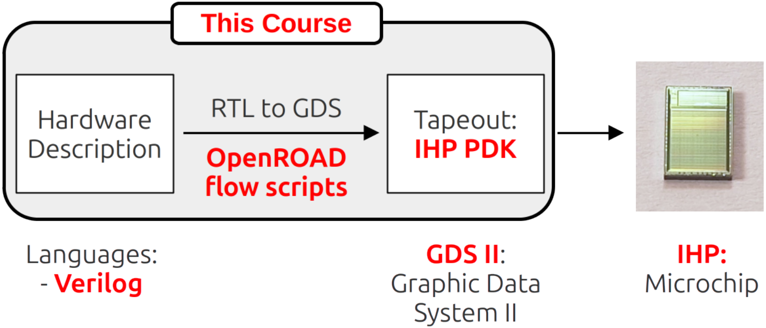
Naming of RTL-to-GDS tools:

The naming of the tools is confusing:

- RTL-to-GDS
- = RTL-2-GDS
- = End-to-End-ASIC tools
- = End-to-End EDA toolchain

They all mean the same.

In this course: ORFS - OpenROAD flow scripts



Many open-Source RTL-to-GDS toolchains

Used with IHP PDK and in this course:

- OpenROAD flow scripts

which is based on

- OpenROAD

Most known other RTL-to-GDS toolchains:

- OpenLANE (based on OpenROAD)
- OpenLANE 2 (based on OpenROAD)
- Silicon Compiler (works with many toolchains, close and open-source)
- Coriolis (developed at University Sorbonne, LIP-6)

A toolchain based on scripts and configuration files

OpenROAD flow scripts are

- based on scripts (obvious in the name)
- based on configuration files

Want most developers know from the commercial tools is:

- Graphical GUIs, used with a mouse and keyboard (shortcuts).
- Configuration through graphical masks, windows, forms.

This might feel uncomfortable at the beginning. But it still has some advantages.

Section 4

About open-source EDA

Advantages of open-source in EDA

- A word by Andrew Kahng (head of OpenROAD) about the relevance of open-source EDA

Andrews slides from the keynote speech at the Chipdesign Network June 2024. As the ucsd server is down, this is a link to the wayback machine:

https://web.archive.org/web/20240609214401/https://vlsicad.ucsd.edu/NEWS24/InnovationKeynote-v6-ACTUAL-DISTRIBUTED.pptx

Slides to have a eye on:

- Slide 06: EDA is an optimization problem
- Slide 10: Open-source EDA and disruptive Innovations
- Slide 13: Open-source accelerates EDA
- The complete chapter “Optimization and Virtuous Cycles” starting at slide 25
- Slides 39-43 about AI and proxies.

Some aspects of open-source EDA

- Three well known PDKs are open-source and production-ready.
- Some other open-source PDKs are not that visible or prominent (MiniFab, Pragmatic(soon?), ...)
- More then one RTL-to-GDS toolchain is production tested.
- Academia starts teaching a lot with open-source EDA.
- Building microchips with open-source became easy and affordable.
- No NDAs, No licence costs, Start with a laptop and internet.

What people have done with open-source EDA

- The following slides contain some works that were made with open-source EDA tools and open-source PDKs.
- All the pictures represent GDS data in different ways.
- A few slides back a picture of GDS in KLayout was shown. The following GDS pictures were created with various other open-source tools (Blender, 3d prints, STL files, . . .).
- Most of these works would not have been possible to integrate here in closed source (because of NDAs and licenses)
- Open-source EDA drives people to experiement and play with the technology.

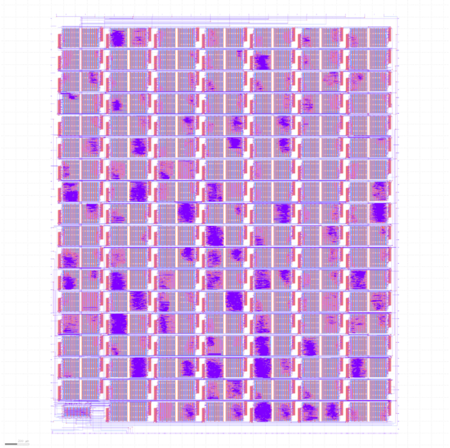


Figure 4: Tinytapeout GDS [3]

[3] Picture by T.Knoll under Creative commons

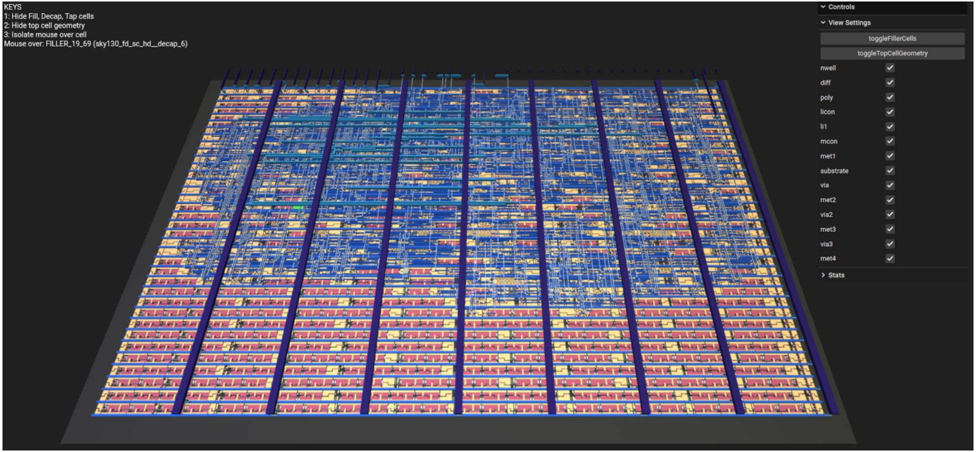


Figure 5: Webviewer Tinytapeout zoomed out [4]

[4]Picture by T.Knoll under Creative commons

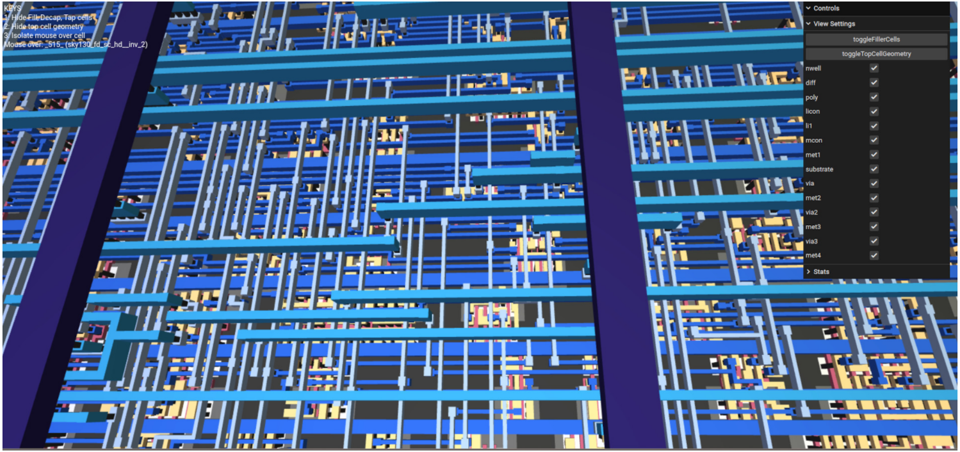


Figure 6: Webviewer Tinytapeout zoomed in [5]

[5] Picture by T.Knoll under Creative commons

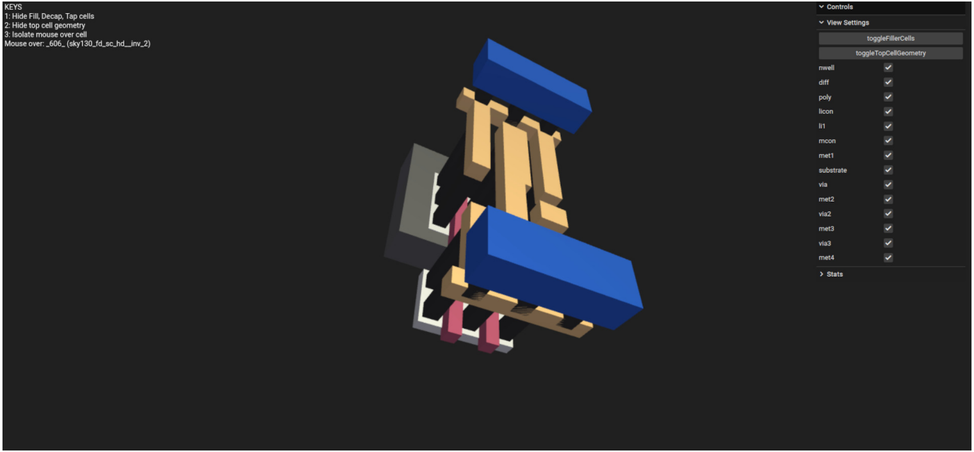


Figure 7: Webviewer Tinytapeout single cell [6]

[6] Picture by T.Knoll under Creative commons

Webviewer of Tinytapout designs with the IHP PDK

See a TinyTapeout design (VGA clock by Matt Venn) in a 3D viewer:

https://tinytapeout.com/runs/ttihp0p2/tt_um_vga_clock

It is made with the IHP open-source PDK.

Tinytapeout Demoscene Competition

Tinytapeout competition to produce a VGA screen and sound demo with around 4000 gates.

https://tinytapeout.com/competitions/demoscene-tt08-winners/

https://www.a1k0n.net/2025/12/19/tiny-tapeout-demo.html

It is made with the SkyWater 130nm open-source PDK.

Siliwiz - How do semiconductors work?

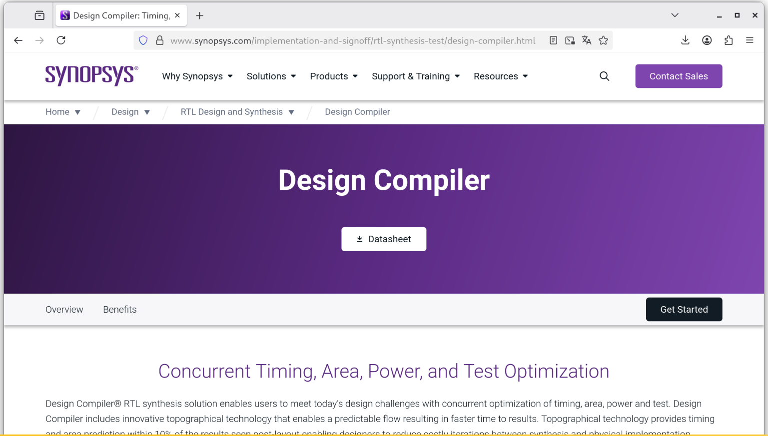
Play with Siliwiz or take the lessons:

https://tinytapeout.com/siliwiz/introduction/

https://app.siliwiz.com/

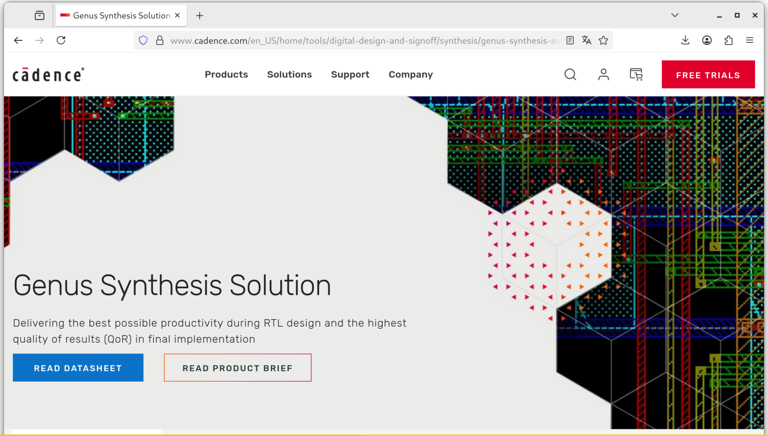
Commercial EDA - Synopsys Design Compiler

https://www.synopsys.com/implementation-and-signoff/rtl-synthesis-test/design-compiler.html



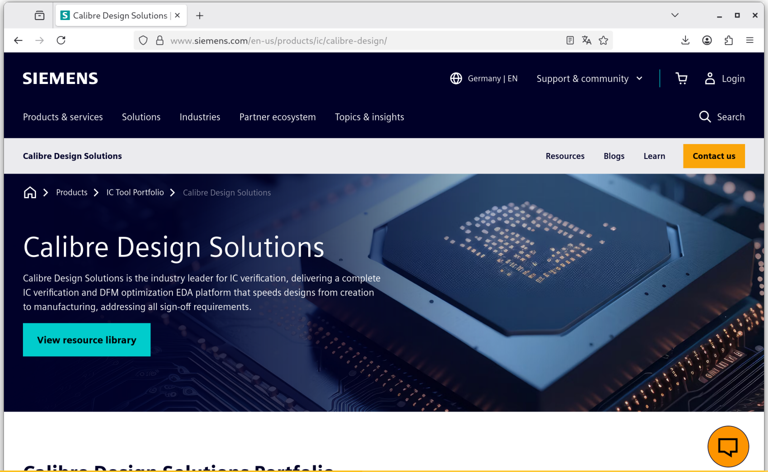
Commercial EDA - Cadence Genus

https://www.cadence.com/en_US/home/tools/digital-design-and-signoff/synthesis/genus-synthesis-solution.html



Commercial EDA - Siemens Calibre DRC/LVS

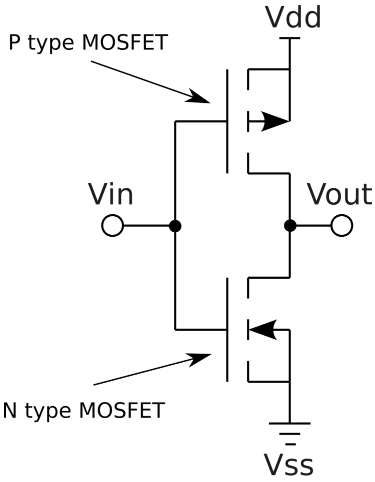
https://www.siemens.com/en-us/products/ic/calibre-design/



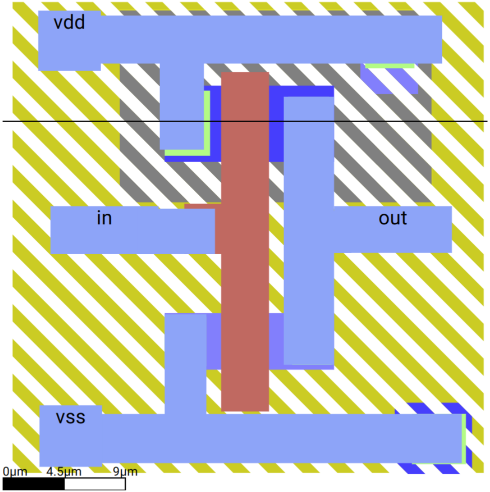
Section 5

CMOS Inverter Example

CMOS Inverter Schematics



CMOS Inverter Layout - SiliWiz simple



Open a real inverter with KLayout tool

Clone the IHP Process Development Kit and open the KLayout gui.

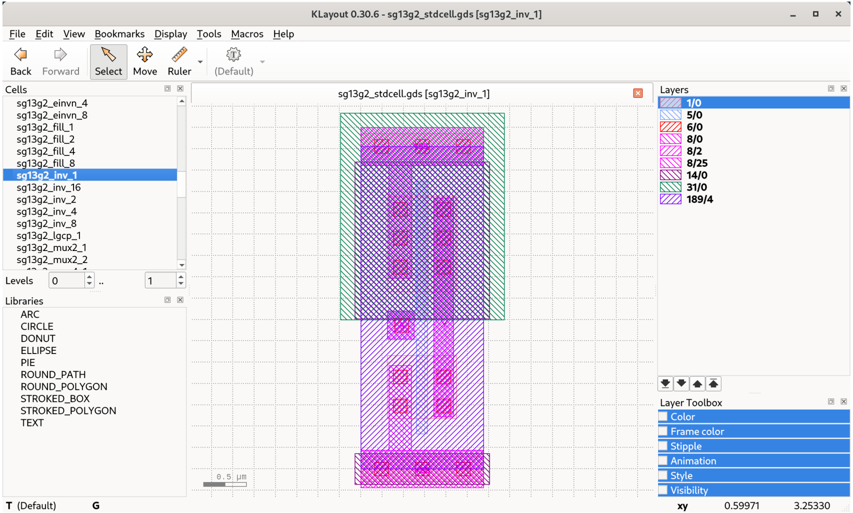
```
git clone https://github.com/IHP-GmbH/IHP-Open-PDK
cd IHP-Open-PDK
klayout
```

Now open the GDS2 file with all library cells for the IHP process via the window menu

```
File->Open (libs.ref/sg13g2_stdcell/gds/sg13g2_stdcell.gds)
```

Select the cell “sg13g2_inv_1” in the cell selection window Right-Click the cell and choose “Show As New Top”

KLayout GDS2 viewer for IHP inverter with layer names



Add Layer Properties

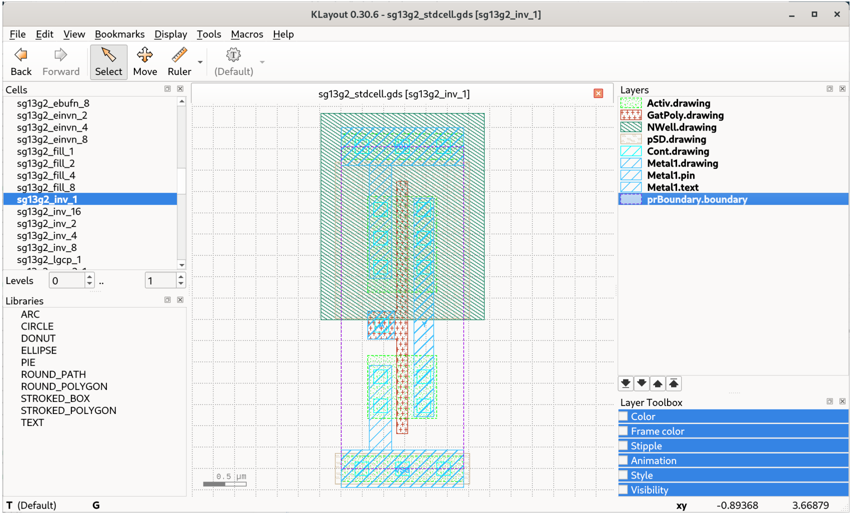
The GDS2 file does not have name definitions for the layers - only numbers. To add the layer definitions go to the menu item

```
File->Load Layer Properties
```

and load

```
libs.tech/klayout/tech/sg13g2.lyp
```

KLayout GDS2 viewer for IHP inverter



Section 6

Fabs

IHP - Leibniz Institute for High Performance Microelectronics

Wikipedia - IHP

SkyWater

Wikipedia - SkyWater

GlobalFoundries

Wikipedia - GlobalFoundries

TSMC - Taiwan Semiconductor Manufacturing Company

Wikipedia - TSMC

Google Maps - TSMC